



# NEXUS ONE SECURITY

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SMART CONTRACT SECURITY AUDIT

## OL Network Review

JULY 2026 · CONFIDENTIAL

# Introduction

A time boxed security review of the **OL Network** protocol was performed by **Nexus One Security**, with a focus on the security aspects of the smart contract implementation. The audit targeted the project smart contract suite with the objective of identifying and mitigating potential vulnerabilities, thereby strengthening the security and robustness of the blockchain infrastructure.

## About OL Network

OL Network is a Move based blockchain protocol. This review focuses on the Lockbox module in the Slow Wallet v2.0 implementation, which allows users to lock assets for a specified period and receive rewards that are gradually unlocked over time through a daily drip mechanism.

## About Nexus One Security

Nexus One Security delivers independent smart contract audits and formal verification, reading code line by line to prove what is broken and remaining a long term security partner after the report is delivered. We secure protocols across Move on Sui and Aptos, Solidity on EVM chains, and Rust on Solana and beyond.

## Disclaimer

A smart contract security review can never verify the complete absence of vulnerabilities. This is a time, resource, and expertise bound effort where we try to find as many vulnerabilities as possible. We cannot guarantee 100% security after the review, nor that the review will find any problems with your smart contracts. Subsequent security reviews, bug bounty programs, and on chain monitoring are strongly recommended.

## Audit Summary

This report outlines the findings from a comprehensive security audit of the OL Network smart contract suite. The audit revealed findings across the severity levels below. Recommendations for remediation are provided to address these vulnerabilities effectively.

|                      |                  |                    |                           |
|----------------------|------------------|--------------------|---------------------------|
| <b>1</b><br>CRITICAL | <b>0</b><br>HIGH | <b>0</b><br>MEDIUM | <b>2</b><br>INFORMATIONAL |
|----------------------|------------------|--------------------|---------------------------|

## Severity Classification

| SEVERITY           | IMPACT: HIGH | IMPACT: MEDIUM | IMPACT: LOW |
|--------------------|--------------|----------------|-------------|
| Likelihood: High   | Critical     | High           | Medium      |
| Likelihood: Medium | High         | Medium         | Low         |

| SEVERITY        | IMPACT: HIGH | IMPACT: MEDIUM | IMPACT: LOW |
|-----------------|--------------|----------------|-------------|
| Likelihood: Low | Medium       | Low            | Low         |

**Impact** the technical, economic, and reputation damage of a successful attack.

**Likelihood** the chance that a particular vulnerability is discovered and exploited.

**Severity** the overall criticality of the risk.

## Scope

► Slow Wallet v2.0 Lockbox module

# Critical Severity

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N1-01Arbitrary Lock Duration Vulnerability in Lockbox ModuleCRITICAL |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| LOCATION                                                             | Slow Wallet v2.0, Lockbox module<br>Function: new                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DESCRIPTION                                                          | <p>The Lockbox module in the Slow Wallet v2.0 implementation allows users to lock assets for a specified period. However, a vulnerability exists where the DEFAULT_LOCK_DURATION constant is not enforced, allowing users to create lockboxes with arbitrary durations. This oversight can be exploited by malicious actors to create lockboxes with very short durations, leading to inflated reward distributions.</p> <p>The reward distribution mechanism is based on the duration of the lockbox. Rewards are calculated using a daily drip mechanism that gradually unlocks assets over time. The daily drip is inversely proportional to the lock duration, so shorter durations result in higher daily rewards.</p> <p><b>Daily drip calculation.</b> The daily drip value is calculated using the formula <code>daily_drip_value = total_locked_value / duration_in_days</code>, where <code>duration_in_days</code> is derived from the lock duration in months.</p> <p><b>Short duration exploit.</b> If a user sets a very short duration such as 1 day, <code>duration_in_days</code> becomes very small, leading to a large <code>daily_drip_value</code>. This results in a higher proportion of the locked value being unlocked daily.</p> <p><b>Reward inflation.</b> By creating multiple lockboxes with short durations, a malicious actor can repeatedly receive high daily rewards, inflating their total rewards significantly compared to the intended design.</p> |
| PROOF OF CONCEPT                                                     | <p>The <code>exploit_short_duration</code> function simulates the creation of a lockbox with a very short duration such as 1 day. This duration is not part of the predefined <code>LOCK_DURATIONS</code>, allowing the attacker to exploit the reward calculation. The <code>test_exploit</code> function sets up a test environment, creates the short duration lockbox, and asserts that the lockbox was created and that the resulting balance is inflated beyond the initial amount.</p> <pre>#[test(framework = @0x1, bob_addr = @0x10002)] fun test_exploit(framework: &amp;signer, bob_addr: address) acquires SlowWalletV2 {   // Setup test environment   let bob_sig = account::create_account_for_test(bob_addr);   let coin = test_setup(framework, 1000); // Mint 1000 coins for testing    // Execute the exploit   exploit_short_duration(&amp;bob_sig, coin);    // Verify the exploit by checking the balance   let (found, idx) = idx_by_duration(bob_addr, 1); // Check for the short duration lockbox   assert!(found, 7357001); // Ensure the lockbox was created   let balance = user_balance(bob_addr);   assert!(balance &gt; 1000, 7357002); // Check if the balance is inflated }</pre>                                                                                                                                                                                                                                                                        |
| RECOMMENDATION                                                       | Enforce the use of predefined lock durations by validating the duration against the <code>LOCK_DURATIONS</code> constant before allowing the creation of a lockbox. This ensures that only valid durations are used, preventing the creation of lockboxes with arbitrary durations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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|            | <pre>public(friend) fun new(locked_coins: Coin&lt;LibraCoin&gt;, duration_type: u64): Lockbox {     // Validate that the duration is in the allowed list     assert!(is_valid_duration(duration_type), error::invalid_argument(EINVALID_DURATION));     let (midnight, _) = date::todays_start_seconds();     Lockbox {         locked_coins,         duration_type,         lifetime_deposited: 0,         lifetime_unlocked: 0,         last_unlock_timestamp: midnight,     } }</pre> |
| RESOLUTION | <b>FIXED</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

DETAILED FINDINGS

# Informational

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>N1-02</b>   | <b>Unused Constant Definitions in Lockbox Module</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>INFO</b> |
| LOCATION       | Slow Wallet v2.0, Lockbox module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| DESCRIPTION    | <p>The Lockbox module declares several constants that are never referenced anywhere in the codebase. Dead constants add noise, can mislead readers about intended behavior, and may hide the fact that a guard was meant to be enforced but never wired up.</p> <pre>const DEFAULT_LOCK_DURATION: u64 = 60;    // never enforced const MAX_LOCKBOXES_PER_USER: u64 = 32;   // never referenced const SECONDS_PER_DAY: u64 = 86400;      // never referenced</pre> <p>Of particular note, <code>DEFAULT_LOCK_DURATION</code> is defined but never used, which is directly related to the arbitrary lock duration issue described in finding N1-01.</p> |             |
| RECOMMENDATION | Remove the unused constants to keep the module clean and maintainable. If a constant such as <code>DEFAULT_LOCK_DURATION</code> or <code>MAX_LOCKBOXES_PER_USER</code> is intended to enforce a limit, wire it into the relevant validation logic rather than leaving it as dead code.                                                                                                                                                                                                                                                                                                                                                                |             |
| RESOLUTION     | <b>FIXED</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |

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| <b>N1-03</b> | <b>Unused Helper Function <code>days_since_epoch</code></b>                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>INFO</b> |
| LOCATION     | Slow Wallet v2.0, Lockbox module<br>Function: <code>days_since_epoch</code>                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| DESCRIPTION  | <p>The private helper function <code>days_since_epoch</code> is defined but is not called anywhere in the module. It appears to be a leftover from an earlier iteration of the daily drip calculation. Unused functions increase the maintenance surface, can confuse auditors about which time computation path is authoritative, and may drift out of sync with the code that is actually used.</p> <pre>fun days_since_epoch(timestamp: u64): u64 {     timestamp / SECONDS_PER_DAY }</pre> |             |

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|                |                                                                                                                                                                                                                          |
| RECOMMENDATION | If the function is intended for future use or as part of a public API, document its purpose and add test coverage. If it is not needed, remove it so that only the authoritative time computation remains in the module. |
| RESOLUTION     | <div>FIXED</div>                                                                                                                                                                                                         |